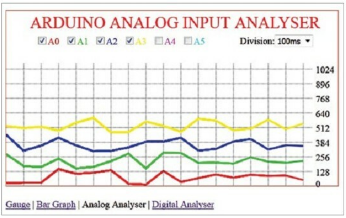

Fig. 6: Analogue input

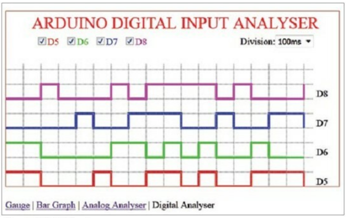

Fig. 7: Digital input

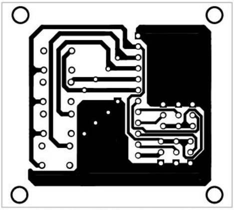

Fig. 8: Actual-size PCB layout for graphical data display with Arduino and HTML5

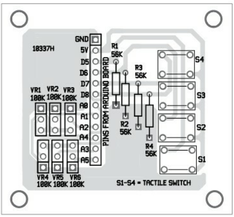

Fig. 9: Components layout for the PCB

pins A0 through A5 of the Arduino. The bar height varies from 0 to 1023 as input varies from 0V to 5V. This type of graph can be used to compare the latest data values with each other. If you require variation in data values with time, the next graph may be used.

Analogue input analyser. Here another type of graph called line graph is used to plot Arduino data variations with respect to time. The lines are formed by connecting points corresponding to data values sampled at specific intervals. The selection of duration between the samples is given in the display itself. Similar to the bar graph, inputs A0 through A5 are used. As inputs can overlap on the graph, these are plotted with unique coloured lines to distinguish them from one another.

Digital input analyser. Digital input analyser display is essentially a modified line graph with data values as 'low' or 'high' only. Instead of changing the value over the duration, it is changed as a step at the instance of change. Hence the waveform looks like a series of pulses. This is helpful in seeing the changes in the binary state of a particular data generated by Arduino. In this project the input state of four digital pins are used as four channels to be monitored. While using the digital pins, care must be taken not to use pins being used by Ethernet shield to communicate with Arduino board.

In this setup we have used digital pins 5 through 8 of Arduino as the inputs. These pins are connected to four pushbutton switches (S1 through S4). Pressing the switch changes the input state from low to high. The canvas graph coded here allows four input channels to be monitored simultaneously.

EFY Note
The source code for this project is included in this month's EFY DVD and is also available for free download at *source.efymag.com*

PARTS LIST	
Semiconductors:	
Board1	- Arduino Uno
Resistors (all 1/4-watt, ±5% carbon):	
R1-R4	- 56-kilo-ohm
VR1-VR6	- 100-kilo-ohm
Miscellaneous:	
S1-S4	- Tactile switch
	- Ethernet shield
	- microSD card

Construction and testing

An actual-size PCB layout for Arduino-based graphical data display is shown in Fig. 8 and its components layout in Fig. 9. After assembling the circuit along with Ethernet shield on the PCB, connect the Arduino Uno to a PC by using a USB cable between the USB port of Arduino Uno and COM port of the PC. Similarly, connect Ethernet shield to the PC using Ethernet cable between their Ethernet ports. After connection, browse the shield's IP address in the browser and see the various parameters.

Further applications

Apart from the aforementioned types of graphs/charts, many more applications are possible. For example, temperature reading can be shown on thermometer graphs and values can be compared using pie charts, area charts, scatter charts, etc. As canvas is essentially a part of HTML5, it works wherever HTML5 is supported. As of now, it is supported in the recent versions of all major browsers including mobile apps. Therefore if the web server is accessible on the Internet, the display can be remotely monitored on a wide range of devices including desktops, tablets and smart phones. EFY


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